

Ming Hei (Calvin) Wong

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EDUCATION

- Ph.D. Mathematics, University of Tennessee, Knoxville, August 2022–present; expected 2027
Co-advisers: Steven M. Wise (UTK) and Cory D. Hauck (ORNL)
- M.S. Mathematics, University of Tennessee, Knoxville, May 18, 2024
- B.S. Mathematics (Computational and Applied Mathematics, Enrichment Mathematics),
First Class Honour, The Chinese University of Hong Kong, July 2022

APPOINTMENTS

- 2024–2026 Oak Ridge National Laboratory
Summer Internship Scientist, June–August 2024, 2025 and 2026
- 2022– University of Tennessee, Knoxville
Graduate Teaching Assistant, 2022–2023
Graduate Teaching Associate, 2023–present
Graduate Research Assistant, 2023–present
- 2021–2022 The Chinese University of Hong Kong
Undergraduate Researcher, Undergraduate Research Opportunity Program, 2021–2022

RESEARCH AREAS

Numerical analysis of PDEs: diffuse-domain methods, asymptotic analysis, adaptive discretizations

Multigrid methods: convergence theory, nonlinear multigrid, block-structured adaptive solvers

Scientific machine learning: graph neural networks, reinforcement learning, combinatorial optimization

Agentic scientific computing: AI-assisted research software with executable verification

Earlier work: mathematical imaging, computational differential and quasiconformal geometry

TECHNICAL SKILLS AND LANGUAGES

Programming and computing: MATLAB, Python, Swift, $\text{\LaTeX}$ /TikZ, Git; remote HPC workflows

Agentic software development with Codex and Claude Code; MCP tool design for AI agents

Languages: English and Cantonese (fluent); Mandarin (basic)

RESEARCH EXPERIENCE

2024– Learning-Guided Combinatorial Optimization for Lattice Alloys

- Oak Ridge National Laboratory and University of Tennessee, Knoxville; with Cory D. Hauck (Ph.D. co-adviser); collaborators Massimiliano Lupo Pasini (ORNL) and Michael S. Murillo (Michigan State University)
- Designed a graph neural network that ranks candidate atom swaps, using species embeddings, graph convolutions and order-invariant pair features; one trained checkpoint is reused across lattice sizes and compositions
- Implemented the physics environment, supervised and double-Q (six-step return) learners, training-data generation, and classical baselines including Metropolis–Hastings, simulated annealing and parallel tempering; ran experiments locally and on HPC clusters
- Benchmarked search-phase performance under equal wall-clock budgets, excluding training, with a Lennard-Jones energy model; drafting the manuscript as planned first author

2022– Diffuse-Domain Methods for PDEs on Complex Geometries

- University of Tennessee, Knoxville; with Steven M. Wise (Ph.D. co-adviser), Tadele Mengesha and Toai Luong
- Wrote MATLAB block-structured adaptive multigrid (BSAM) solvers for diffuse-domain problems, used to test asymptotic predictions on nontrivial geometries
- Designed convergence studies with manufactured solutions, successive-solution comparisons across different adaptive grid hierarchies, and regression on candidate asymptotic terms
- Numerically investigated coupled interface-width and regularization refinement paths, which motivated a formal second-order analysis; drafted and organized the resulting joint manuscript (in preparation)

2026 Symmetric V-Cycle Convergence under Cell-Centred Multigrid Axioms

- Proved, as sole author, that the cell-centred multigrid axioms do not imply symmetric V-cycle convergence: constructed hierarchies with V-cycle spectral radius above one while the W-cycle remains uniformly contractive
- Built an agent-assisted search in which candidate hierarchies were tested with an executable multigrid solver before proof

2024–2025 Multigrid Methods Textbook

- Coauthored with Abner J. Salgado and Steven M. Wise (De Gruyter, 2025)
- Developed the book’s MATLAB codes, diagrams and exposition, including a one-dimensional nonlinear (FASD) implementation

2022 High-Dimensional Numerical Integration with Multilevel Dimension Iteration Methods

- NSF-funded summer research supervised by Prof. Feng Xiaobing, University of Tennessee, Knoxville
- Performed dimension reduction for high-dimensional integration via the co-area formula and Lebesgue measure; conducted numerical experiments on the proposed methods

2021 Detecting Pavement Defects with Mathematical Imaging

- Supervised by Prof. Ronald Lui, The Chinese University of Hong Kong; coordinated a team of six undergraduate and full-time researchers
- Detected potholes from curvature and Beltrami coefficients, and cracks from high-frequency Fourier components; denoised images with total-variation methods and Gaussian smoothing

2021 Classifying Childhood Obstructive Sleep Apnea with Machine Learning and Quasiconformal Geometry

- Supervised by Prof. Ronald Lui, The Chinese University of Hong Kong; coordinated three undergraduate researchers
- Developed a classification model for early-stage diagnosis from 3D face models reconstructed from smartphone photographs; presented the work to an audience of about 100

2019 Mathematical Proof Judging with Natural Language Processing

- Supervised by Dr. Cheung Leung Fu and Dr. Chan Ping Shun, The Chinese University of Hong Kong
- Researched AI-driven proof-judging systems; presented basic algorithms and a prototype for standardized problems

PUBLICATIONS

Book

- 2025 A. J. Salgado, S. M. Wise, and C. Wong. *Multigrid Methods: Axiomatic Convergence Theory for Linear and Weakly Nonlinear Problems*. De Gruyter, Berlin/Boston, 2025 (published online November 2025; © 2026). ISBN 978-3-11-135419-4. <https://doi.org/10.1515/9783111354880>

Journal Articles

- 2025 T. Luong, T. Mengesha, S. M. Wise, and M. H. Wong. “A Diffuse Domain Approximation with Transmission-Type Boundary Conditions II: Γ -Convergence.” *International Journal of Numerical Analysis and Modeling*, 22(5): 728–744, 2025. <https://doi.org/10.4208/ijnam2025-1031>
- 2025 T. Luong, T. Mengesha, S. M. Wise, and M. H. Wong. “A Diffuse Domain Approximation with Transmission-Type Boundary Conditions I: Asymptotic Analysis and Numerics.” *International Journal of Numerical Analysis and Modeling*, 22(5): 694–727, 2025. <https://doi.org/10.4208/ijnam2025-1030>
- 2023 M.-H. Wong, M. Li, K.-M. Tam, H.-M. Yuen, C.-T. Au, K. C.-C. Chan, A. M. Li, and L.-M. Lui. “A Quasiconformal-Based Geometric Model for Craniofacial Analysis and Its Application.” *Axioms*, 12(4): 393, 2023. <https://doi.org/10.3390/axioms12040393>

Preprints

- 2026 T. Luong, T. Mengesha, K. Stinson, S. M. Wise, and M. H. Wong. “The α -Limit Problem: Convergence of a Linear Degenerate Interface Transmission Problem.” arXiv:2609.01237 [math.AP], 2026. <https://arxiv.org/abs/2609.01237>
- 2026 M. H. Wong. “The Symmetric V-Cycle Can Diverge under the Multigrid Axioms for Cell-Centred Discretisations.” arXiv:2607.23391 [math.NA], 2026. <https://arxiv.org/abs/2607.23391>

Manuscripts in Preparation

M. H. Wong et al. “Learning to Rank Candidate Swaps for Zero-Temperature Lattice Configuration Optimization.” Planned first author; internal review nearing completion.

T. Luong, T. Mengesha, S. M. Wise, and M. H. Wong. “Coupled One-Sided Diffuse Domain Approximation: Power-Law and Logarithmic Refinement Paths.”

TALKS AND CONFERENCES

- 2025 UARK–SIAM–CSS 2025 Conference, University of Arkansas, Fayetteville, AR
Invited talk: “Gamma-Convergence and Asymptotic Analysis for a Diffuse Domain Problem with Transmission Boundary Conditions: Part 2, Numerical Confirmation”
- 2025 SIAM SEAS 2025, March 2025
Invited talk: “Gamma-Convergence and Asymptotic Analysis for a Diffuse Domain Problem with Transmission Boundary Conditions: Part 2, Numerical Confirmation”
- 2024 Workshop on Theoretical and Numerical Challenges in Materials Science, The University of Alabama
- 2022 Undergraduate Research Opportunity Program 2021 Workshop, The Chinese University of Hong Kong
Talk: “Preliminary Diagnosis of Childhood Obstructive Sleep Apnea using 2D Images by Quasi-Conformal Geometry”

SOFTWARE

- TexMacgic Native macOS L^AT_EX editor for scientific writing, designed and written in Swift. Includes project and asset tools, Git status, visual TikZ, symbol and table tools, and an integrated agent that works through Model Context Protocol (MCP) tools. Version 1.0.0 released on the Mac App Store, September 2026; version 1.0.1 approved for the Mac App Store:
<https://apps.apple.com/us/app/texmacgic/id6804365513>
- BSAM Open-source MATLAB block-structured adaptive FAS multigrid for 2D elliptic, parabolic and diffuse-domain problems, implementing the method of Feng, Guo, Lowengrub and Wise (2018):
<https://github.com/Calvin1127XD/BSAM-in-MATLAB>
- Multigrid MATLAB codes accompanying the multigrid textbook:
<https://github.com/stevenmwise/MultigridCourse>

TEACHING

University of Tennessee, Knoxville

- 2026 Calculus II (teaching assistant)
- 2025 Finite Mathematics (teaching assistant)
- 2025 Mathematics for the Life Sciences (instructor of record)
- 2024 ODE I (teaching assistant)
- 2024 Multigrid Methods (teaching assistant)

2024	Calculus II (teaching assistant)
2023	Basic Calculus (instructor of record)
2023	Finite Mathematics (teaching assistant)
2022	College Algebra (teaching assistant)

The Chinese University of Hong Kong

2022	Science Academy for Young Talent: A Trilogy of Hands-on Machine Learning (teaching assistant)
2021	Enrichment Programme for Young Mathematics Talent: Introduction to Discrete Mathematics (teaching assistant)
2020	Enrichment Programme for Young Mathematics Talent: Towards Differential Geometry (teaching assistant)
2019	Enrichment Programme for Young Mathematics Talent: Towards Modern Algebra (teaching assistant)

AWARDS, SCHOLARSHIPS AND FELLOWSHIPS

2026	Spike Tickle STEM Endowed Fellowship, University of Tennessee, Knoxville
2025	Dawn and Lawrence Taylor Graduate Fellowship, University of Tennessee, Knoxville
2025	Dorothea & Edgar D. Eaves Graduate Student Teaching Award: Senior, University of Tennessee, Knoxville
2022	Department of Mathematics Fellowship, University of Tennessee, Knoxville
2022	Undergraduate Research Opportunity Program Gold Award, The Chinese University of Hong Kong
2022	Chung Chi College Departmental Prize – Enrichment Mathematics, The Chinese University of Hong Kong
2021	Chung Chi College Scholarship for Excellence, The Chinese University of Hong Kong
2019	Dr Chao Yung Chi-hsing Scholarship in Mathematics 2019/20
2017–22	Faculty of Science Dean’s List, The Chinese University of Hong Kong (5 times)
2017–22	Chung Chi College Kunkle and Pommerenke Full Tuition Scholarship, The Chinese University of Hong Kong (3 times)
2017–22	Undergraduate Mathematics Scholarship, Department of Mathematics, The Chinese University of Hong Kong (5 times)

Updated October 2026